1 Solution — GIAM 1.1 Problem 3

1.1 Problem

Identify each as rational or irrational.

- **(a)** $5021.2121212121\ldots$
- **(b)** $0.2340000000\ldots$
- **(c)** $12.31331133311133331111\ldots$
- **(d)** π
- **(e)** $2.98765432198765432198765 4321\ldots$

1.2 The Key Principle

The textbook hinted at the rule on p.5: *a real number is rational if and only if its decimal expansion is eventually periodic* — i.e. some finite block of digits repeats forever from some point onward. Equivalently, a decimal expansion that *terminates* (ends in all zeros) is rational; an expansion with a repeating block of fixed length is rational; an expansion with *no* eventually-repeating pattern is irrational.

We apply this principle to each item.

1.3 Part (a) — $5021.2121212121\ldots$ — Rational

The decimal point splits this into integer part 5021 and fractional part $0.212121\ldots$ — the digits "21" repeating forever. **Eventually periodic $\Longrightarrow$ rational.**

To convert: let $x = 5021.2121\overline{21}$. Then $100x = 502121.2121\overline{21}$, and subtracting cancels the repeating tail:

$$
\begin{aligned}
100x - x &= 502121 - 5021 = 497100,\\
99\,x &= 497100,\\
x &= \frac{497100}{99} = \frac{165700}{33}.
\end{aligned}
$$

Equivalently, $x = 5021 + \frac{7}{33}$.

1.4 Part (b) — $0.234000000\ldots$ — Rational

The decimal terminates after three digits: the rest of the expansion is all zeros. A terminating decimal is rational by inspection:

$$0.234 = \frac{234}{1000} = \frac{117}{500}.$$

("Terminating" *is* a kind of eventual periodicity: the repeating block is just the single digit "0".)

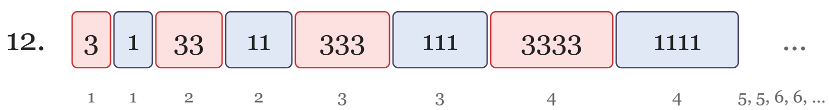
1.5 Part (c) — $12.313311333111333331111\ldots$ — Irrational

Parse the digits after the decimal point:

$$\underbrace{3}_{1}\ \underbrace{1}_{1}\ \underbrace{33}_{2}\ \underbrace{11}_{2}\ \underbrace{333}_{3}\ \underbrace{111}_{3}\ \underbrace{3333}_{4}\ \underbrace{1111}_{4}\ \underbrace{33333}_{5}\ \underbrace{11111}_{5}\ \ldots$$

The pattern: alternating runs of **3s** and **1s**, where the runs **grow longer by one** every two blocks (sizes $1, 1, 2, 2, 3, 3, 4, 4, \ldots$).

Block structure of (c): alternating 3-runs and 1-runs of ever-growing length



block sizes (below each block): 1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 6, 6, ...
— sizes grow without bound, so no finite pattern can repeat forever

3-runs 1-runs

Figure 1.1: Alternating 3-runs (pink) and 1-runs (blue) with sizes 1, 1, 2, 2, 3, 3, 4, 4, ... The block sizes grow without bound, so no finite repeating pattern is possible.

This is *not* eventually periodic. Suppose, toward contradiction, that beyond some position N the digits repeated with some fixed period p. Beyond position N the expansion contains 3-runs of every length $k \geq K$ for some K — in particular, runs longer than p. But within any single period of length p, there can be at most p consecutive $\mathbf{3}$s — so a run of $\mathbf{3}$s of length $> p$ couldn't fit. Contradiction.

So **(c) is irrational**.

1.6 Part (d) — π — Irrational

π is the textbook's running example of an irrational number, and its decimal expansion

$$\pi = 3.14159\,26535\,89793\,23846\,26433\,83279\,\ldots$$

has no observable repeating pattern. The irrationality of π is a classical theorem (first proved by Lambert in 1761; the cleanest modern proof is Niven's 1947 calculus-based argument). The proof is well beyond Chapter 1 — but the result is famous enough to take on faith here. **Irrational.**

1.7 Part (e) — $2.987654321987654321987654321\ldots$ — Rational

The digits after the decimal point repeat the nine-digit block "987654321" forever. **Eventually periodic $\Longrightarrow$ rational.**

To convert: let $x = 2.\overline{987654321}$. Then $10^9 x = 2987654321.\overline{987654321}$, and subtracting cancels the repeating tail:

$$\begin{aligned} 10^9\,x - x &= 2987654321 - 2 = 2987654319, \\ 999999999\,x &= 2987654319, \\ x &= \frac{2987654319}{999999999}. \end{aligned}$$

1.8 Summary

Part	Value	Verdict	Reason
(a)	$5021.212\overline{121}$	rational	Pre-period “5021.”, repeating block “21”.
(b)	$0.234000\ldots$	rational	Terminating decimal $= 117/500$.
(c)	$12.313311333111333331111\ldots$	irrational	Block sizes $1, 1, 2, 2, 3, 3, \ldots$ grow without bound; no fixed period.
(d)	π	irrational	Classical theorem (Lambert 1761).
(e)	$2.\overline{987654321}$	rational	Nine-digit repeating block.

Table 1.1

1.9 Remark — Why “Eventually Periodic $\Leftrightarrow$ Rational”

The forward direction is the construction we used above. Given a decimal x with eventual period p starting at some point, write $10^{a+p}x - 10^{a}x$ where a is the length of the pre-period; the two expansions cancel everything past position a, leaving a finite *integer*. So $x \cdot (10^{a+p} - 10^{a})$ is an integer, making x rational.

The reverse direction comes from the *long division* algorithm. Dividing an integer by another integer q produces, at each step, a remainder in $\{0, 1, \ldots, q-1\}$ — only q possible values. After at most q steps the remainders must repeat (pigeonhole), and once a remainder repeats the digits start cycling. Hence the decimal is eventually periodic with period $\leq q$.

So the equivalence

$$x \in \mathbb{Q} \quad \Longleftrightarrow \quad \text{decimal expansion of } x \text{ is eventually periodic}$$

is genuine, two-sided, and provable elementary.

1.10 Remark — Terminating Decimals Are a Special Case

A terminating decimal like 0.234 can be viewed as $0.234000\overline{0}$ — periodic with the trivial repeating block "0". This is why part (b) fits seamlessly into the "rational $\iff$ eventually periodic" framework.

Some authors write a terminating decimal *two* ways: $0.234 = 0.234000\ldots = 0.233999\ldots$ (the latter using "9" as the repeating block). Both representations are legitimate — and both are eventually periodic, hence rational.

1.11 Remark — How to "Cook Up" an Irrational Like (c)

Item (c) is *engineered* to be irrational by sabotaging periodicity. The recipe:

1. Pick two distinct digits, $d_1 \neq d_2$.
2. Write block-runs of these digits with *increasing* run lengths: $d_1, d_2, d_1d_1, d_2d_2, d_1d_1d_1, d_2d_2d_2, \ldots$
3. Since the run lengths grow without bound, no finite period p can fit *any* sufficiently late long run inside a single period — periodicity fails.

This is a *constructive* way to write down a specific irrational, without invoking $\sqrt{2}$, π, or any other transcendental. Variants include:

- **Champernowne's constant.** $0.123456789101112131415\ldots$ — concatenate all positive integers. Irrational (in fact transcendental).
- **Liouville's constant.** $0.110001000000000000000001\ldots$ — a 1 in positions $1!, 2!, 3!, \ldots$ and 0 elsewhere. The "spacing" grows so fast that the decimal can't be periodic — and in fact it's not even algebraic.

These are all examples of "irrationality by construction" rather than by classical proof.

1.12 Remark — π's Irrationality is Genuinely Hard

The other four parts have *elementary* proofs: just look at the decimal pattern, or run long-division backwards. π is the exception — its decimal expansion looks "patternless" but proving there's no eventual periodicity requires real analysis.

The standard proof goes by way of the integral

$$\int_0^1 x^n(1-x)^n \cdot \sin(\pi x)\, dx,$$

which — assuming $\pi = a/b$ rational — turns out to be both a positive integer (for some large n) and strictly less than 1. Contradiction.

This is *much* harder than the long-division argument that classified (a), (b), (e). The textbook explicitly defers the proof — but the irrationality is so well-known that we accept it as background.

1.13 Remark — A Sneaky Observation about (a) and (e)

Items (a) and (e) are both built from the same *recipe*: pick a non-trivial repeating block, attach an integer part. The size of the repeating block determines the denominator:

- (a) has period 2, so the denominator divides $99 = 10^2 - 1$.
- (e) has period 9, so the denominator divides $999999999 = 10^9 - 1$.

In general, a *purely* periodic decimal with period p is rational with denominator dividing $10^p - 1$. This is exactly why long division of any integer by $10^p - 1$ yields the corresponding repeating decimal. (Try $1/7 = 0.\overline{142857}$ — period 6, and $7 \mid 999999$.)

Mixed pre-period + period (like (a)) just shifts the integer part. The repeating-block-to-denominator relation still controls the fractional part.